

$D(N)$ -QUADRUPLES IN UPPER-TRIANGULAR 2×2 INTEGER MATRICES

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ABSTRACT. We introduce analogues of Diophantine $D(N)$ - m -tuples in the noncommutative ring $M_2(\mathbb{Z})$ of 2×2 integer matrices. Besides definitions based on the standard matrix product, we consider a symmetric version defined via the Jordan product

$$A \circ B = \frac{1}{2}(AB + BA).$$

Special attention is devoted to upper-triangular integer matrices $UT_2(\mathbb{Z})$, where squares admit a particularly simple description. Motivated by the classical connection between representations of n as a difference of two squares and the existence of $D(n)$ -quadruples in commutative rings, we investigate the existence of Jordan $D(N)$ -quadruples in $UT_2(\mathbb{Z})$.

1. INTRODUCTION AND MOTIVATION

As a supplement to the book [9] a list of open problems was presented. In this paper, we address the specific problem formulated as follows:

Define an analogue of a $D(n)$ - m -tuple in a noncommutative ring and prove some non-trivial results concerning the existence of $D(n)$ -quadruples, e.g. in the ring of 2×2 matrices with integer entries.

Recall that $D(n)$ - **m -tuples** are traditionally defined in a commutative ring $\mathcal{R}$ with unity as a set of m distinct, nonzero elements $\{a_1, \dots, a_m\}$ in $\mathcal{R}$ such that the product of any two distinct elements, increased by n is a perfect square in $\mathcal{R}$, i.e. $a_i a_j + n = x_{ij}^2$ for all $1 \leq i < j \leq m$, where $n \in \mathcal{R} \setminus \{0\}$.

So far $D(n)$ - m -tuples have been extensively investigated in various rings of integers of number fields, such as $\mathbb{Z}, \mathbb{Z}[\sqrt{d}], \mathbb{Z}[(1 + \sqrt{d})/2]$, as well as in rings of polynomials with integer coefficients. Interestingly, in many of these rings, an equivalence has been established between the representability of n as a difference of two squares and the existence of a $D(n)$ -quadruple (see [6, 8, 12–19]). Yet, this “rule” is not without its counterexamples. In certain rings of some quadratic fields, there exist elements n that cannot be represented as a difference of two squares, yet $D(n)$ -quadruples still exist ([2, 3, 5]).

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A classical construction shows that every $D(n)$ -pair $\{a, b\}$ satisfying $ab + n = r^2$ can be extended to a **regular $D(n)$ -triple**

$$\{a, b, c_{\pm}\}, \quad c_{\pm} = a + b \pm 2r$$

provided that the chosen element $c_{\pm}$ does not belong to $\{0, a, b\}$. Following this construction, one can consider sets of the form:

$$\{a, b, a + b \pm 2r, a + 4b \pm 4r\}.$$

These sets are $D(n)$ -quadruples if the condition $a(a + 4b + 4r) + n = \square$ is satisfied and if elements are nonzero and distinct. Building on this approach, Dujella [7] derived several parametric formulas for $D(n)$ -quadruples. For instance, the set

$$(1) \quad \{m, m(3k + 1)^2 + 2k, m(3k + 2)^2 + 2k + 2, 9m(2k + 1)^2 + 8k + 4\}$$

is shown to be a $D(n)$ -quadruple with $n = 2m(2k + 1) + 1$, assuming the elements are mutually distinct and nonzero. These polynomial families illustrate that for infinitely many values of n , $D(n)$ -quadruples exist in abundance within commutative rings.

Furthermore, an important "scaling property" of these sets is as follows: if the set $\{a_1, \dots, a_m\}$ is a $D(n)$ - m -tuple in a commutative ring $\mathcal{R}$, then for any $w \in \mathcal{R} \setminus \{0\}$, the set

$$\{a_1 w, \dots, a_m w\} \text{ is a } D(nw^2)\text{-}m\text{-tuple.}$$

The aim of the present paper is to initiate the study of Diophantine $D(N)$ -tuples in a genuinely noncommutative setting. We introduce several natural analogues of $D(N)$ -tuples in $M_2(\mathbb{Z})$, with particular emphasis on the symmetric definition induced by the Jordan product. Our main results concern the ring $UT_2(\mathbb{Z})$ of upper-triangular 2×2 integer matrices. Using the characterization of upper-triangular matrices representable as differences of two squares obtained in [11], we prove that every matrix $N \in UT_2(\mathbb{Z})$ representable as a difference of two squares in $UT_2(\mathbb{Z})$ admits infinitely many Jordan $D(N)$ -quadruples (Theorem 5 and Corollary 6). Thus, to the best of our knowledge, $UT_2(\mathbb{Z})$ provides the first structure in which the implication

$$N = K^2 - A^2 \implies \text{there exists a Jordan } D(N)\text{-quadruple}$$

can be established by combining a structural characterization of matrices representable as differences of two squares with an explicit construction of Jordan $D(N)$ -quadruples.

This approach differs substantially from earlier investigations of Diophantine quadruples in rings of algebraic integers (see [12–17]), where existence was typically obtained through explicit polynomial constructions (using (1), for example). We also investigate matrices N that are not representable as differences of two squares. In several infinite families we prove that Jordan $D(N)$ -quadruples do not exist by means of modular obstructions (Propositions 7, 8, 11, 14, 15, 17 and 18), while Propositions 9, 10, 13 and 16 provide explicit constructions of Jordan $D(N)$ -quadruples for matrices belonging to

exceptional congruence classes. These results demonstrate that the relationship between difference-of-squares representations and the existence of Jordan $D(N)$ -quadruples in matrix rings might be more subtle than in the classical commutative setting.

2. MATRIX ANALOGUES OF $D(N)$ - m -TUPLES

Let $M_2(\mathbb{Z})$ be the ring of all 2×2 matrices with integer entries. It is well known that $M_2(\mathbb{Z})$ forms a noncommutative ring with unity I . Due to the lack of commutativity, the standard definition of a Diophantine m -tuple must be adapted. We propose two primary frameworks for defining a $D(N)$ - m -tuple in this context, where $N \in M_2(\mathbb{Z})$.

2.1. Standard Product Definitions. The first approach considers the order of multiplication directly.

Definition 1. Let $N \in M_2(\mathbb{Z})$.

An ordered family $(A_1, \dots, A_m)$ of distinct nonzero matrices in $M_2(\mathbb{Z})$ is called:

- a $D(N)$ **matrix m -tuple in ascending order** if for all $1 \leq i < j \leq m$, there exists $X_{ij} \in M_2(\mathbb{Z})$ such that $A_i A_j + N = X_{ij}^2$,
- a $D(N)$ **matrix m -tuple in descending order** if for all $1 \leq i < j \leq m$, there exists $X_{ji} \in M_2(\mathbb{Z})$ such that $A_j A_i + N = X_{ji}^2$,
- a $D(N)$ **matrix m -tuple** if it is a $D(N)$ matrix m -tuple in both ascending and descending order.

Note that if a m -tuple $(A_1, \dots, A_m)$ consists of matrices that pairwise commute, these three definitions collapse into the classical definition.

Example 1. Let $a \in \mathbb{Z}$ and

$$A = \begin{pmatrix} a+1 & a+2 \\ a & a+1 \end{pmatrix}, \quad B = \begin{pmatrix} -a-1 & a+1 \\ a & -a \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 2a+3 \\ 0 & -2a-1 \end{pmatrix}.$$

Note that

$$\begin{aligned} AB + I &= \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}^2, \quad BA + I = \begin{pmatrix} a & a+1 \\ -a & -a-1 \end{pmatrix}^2, \\ AC + I &= \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}^2, \quad CA + I = \begin{pmatrix} (a+1)(2a+1) & 2a^2+5a+3 \\ -a(2a+1) & -a(2a+3) \end{pmatrix}^2, \\ BC + I &= \begin{pmatrix} 1 & -2(a+1) \\ 0 & 2a+1 \end{pmatrix}^2, \quad CB + I = \frac{1}{4(1+a)^2} \begin{pmatrix} 2a^2+5a+2 & -a(2a+3) \\ -a(2a+1) & 2a^2+3a+2 \end{pmatrix}^2 \\ \text{and } CB + I &= \begin{pmatrix} 0 & 1 \\ 3 & -2 \end{pmatrix}^2, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ for } a = -2, 0, \text{ respectively.} \end{aligned}$$

Thus, for $a \in \{-2, 0\}$, (A, B, C) is a $D(I)$ -triple in the sense of the standard product (in both ascending and descending orders), despite the fact that the matrices do not commute.

2.2. The Jordan Product Approach. As an alternative, we can utilize the symmetric part of the matrix product to restore commutativity at the operational level. We define the Jordan product of two matrices A and B as:

$$A \circ B = \frac{1}{2}(AB + BA),$$

see, for example, [20]. If $\mathbb{F}$ is a field of characteristic different from 2, then $(M_2(\mathbb{F}), \circ)$ is a Jordan algebra. This operation is commutative but generally non-associative, satisfying the Jordan identity: $(A \circ B) \circ A^2 = A \circ (B \circ A^2)$. Using this operation, we can provide a definition that is inherently symmetric:

Definition 2. *The set $\{A_1, A_2, \dots, A_m\}$ of distinct matrices in $M_2(\mathbb{Z})$ is a $D(N)$ **matrix m -tuple under the Jordan product** if for all $1 \leq i < j \leq m$, there exists $X_{ij} \in M_2(\mathbb{Z})$ such that:*

$$(2) \quad A_i \circ A_j + N = X_{ij}^2.$$

Throughout the paper we refer to these simply as **Jordan $D(N)$ - m -tuples**.

A family of matrices $(A_1, A_2, \dots, A_m)$ in $M_2(\mathbb{Z})$, whose elements need not be distinct, is said to have the **Jordan $D(N)$ -property** if it satisfies (2) for all $1 \leq i < j \leq m$.

Although the Jordan product ensures symmetry in the $D(N)$ - m -tuple definition, a potential issue arises as $A \circ B$ may not have integer entries for all pairs $A, B \in M_2(\mathbb{Z})$. Despite this constraint, the Jordan product offers a significant advantage regarding the extension of $D(N)$ -tuples. In this setting, the classical extension formula for triples remains valid. If $\{A, B\}$ is a $D(N)$ pair such that $A \circ B + N = R^2$, then the set

$$\{A, B, A + B \pm 2R\}$$

forms a $D(N)$ -triple. Indeed, bilinearity of the Jordan product allows for the following:

$$A \circ (A + B \pm 2R) + N = A^2 + A \circ B \pm 2(A \circ R) + N = A^2 + R^2 \pm 2(A \circ R) = (A \pm R)^2.$$

Unlike in commutative integral domains, the ring $M_2(\mathbb{Z})$ contains zero divisors which allow for the construction of $D(N)$ m -tuples of arbitrary length.

Example 2. *Let*

$$A_i = \begin{pmatrix} 0 & k_i \\ 0 & 0 \end{pmatrix}, \quad k_i \in \mathbb{Z}.$$

*Note that $A_i A_j = A_j A_i = A_i \circ A_j = \mathbf{0}$ (zero matrix), so $A_i A_j + N^2 = N^2$ for any $N \in M_2(\mathbb{Z})$. Thus, the set $\{A_i\}$ constitutes a $D(N^2)$ -sequence or m -tuple of **arbitrary length** in both the standard and Jordan products.*

Considering Example 2, natural questions arise. How should such degenerate cases be excluded? Should one restrict attention to matrices with

nonzero determinant, or require that the product (or Jordan product) of any two matrices in a $D(N)$ -tuple be nonzero?

Indeed, Example 2 shows that the existence problem becomes trivial if no additional restrictions are imposed. Therefore, throughout the paper we distinguish such degenerate examples from non-degenerate constructions, in which all pairwise Jordan products are nonzero.

Definition 3. A Jordan $D(N)$ - m -tuple is called **non-degenerate** if $A_i \circ A_j \neq 0$, for all $i \neq j$. Otherwise, it is called **degenerate**.

Lemma 1. Let $\{A_1, \dots, A_m\}$ be a Jordan $D(N)$ -tuple in $M_2(\mathbb{Z})$. If $E \in GL_2(\mathbb{Z})$ is a unimodular matrix, then $\{EA_1E^{-1}, \dots, EA_mE^{-1}\}$ is a Jordan $D(ENE^{-1})$ - m -tuple in $M_2(\mathbb{Z})$.

Proof. For every $1 \leq i < j \leq m$, we have

$$A_i \circ A_j + N = X_{ij}^2, \quad X_{ij} \in M_2(\mathbb{Z}).$$

By multiplying the previous relation by E and E^{-1} , we get

$$\frac{1}{2}(E(A_iA_j + A_jA_i)E^{-1}) + ENE^{-1} = EX_{ij}^2E^{-1},$$

$$\frac{1}{2}((EA_iE^{-1})(EA_jE^{-1}) + (EA_jE^{-1})(EA_iE^{-1})) + ENE^{-1} = (EX_{ij}E^{-1})^2.$$

Therefore,

$$(EA_iE^{-1}) \circ (EA_jE^{-1}) + ENE^{-1} = (EX_{ij}E^{-1})^2.$$

Because E and E^{-1} are unimodular integer matrices, all matrix products have integer entries, and $(EA_iE^{-1}) \circ (EA_jE^{-1})$ as well. $\square$

Example 3. Conjugating the quadruple from Example 4 by the unimodular parametric matrix:

$$E = \begin{pmatrix} 1 & k \\ m & 1 + km \end{pmatrix}, \quad k, m \in \mathbb{Z},$$

we obtain $D(N_m)$ -quadruple for $N_m = \begin{pmatrix} 2(2-m) & 2 \\ -2m^2 & 2(2+m) \end{pmatrix}$:

$$\left\{ \begin{pmatrix} -m & 1 \\ -m^2 & m \end{pmatrix}, \begin{pmatrix} -2(1+6k)m & 2(1+6k) \\ -2m(6+m+6km) & 2(6+m+6km) \end{pmatrix}, \right. \\ \left. \begin{pmatrix} 3-2m+2km & 2(1-k) \\ 2m(1-m+km) & 1+2m-2km \end{pmatrix}, \begin{pmatrix} 7-m+2km & 1-2k \\ m(2-m+2km) & 5+m-2km \end{pmatrix} \right\}$$

3. UPPER-TRIANGULAR JORDAN $D(N)$ -QUADRUPLES

3.1. The $UT_2(\mathbb{Z})$ setting. A natural setting in which to study the existence of $D(N)$ -matrix tuples is the ring of upper-triangular 2×2 integer matrices, matrices of the form

$$T = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \quad a, b, c \in \mathbb{Z}.$$

We denote this set by $UT_2(\mathbb{Z})$. Note that $UT_2(\mathbb{Z})$ forms a ring under standard addition and matrix multiplication.

The Jordan product is not closed in $UT_2(\mathbb{Z})$ since

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \circ \begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & \frac{1}{2}(e(a+c) + b(d+f)) \\ 0 & cf \end{pmatrix}.$$

Let us mention that the structure $(UT_2(\mathbb{Q}), +, \circ)$ forms a commutative but non-associative algebra, known as a *special Jordan algebra*. Despite this, the set $UT_2(\mathbb{Z})$ is a convenient setting for constructing explicit examples of Jordan $D(N)$ - m -tuples, because squares in this ring have a very simple form:

$$(3) \quad T^2 = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix}.$$

From now on, $UT_2(\mathbb{Z})$ is our ambient structure.

The following lemma shows that the existence of a Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$ for $N \in UT_2(\mathbb{Z})$ whose upper-right entry is 1 implies existence of Jordan quadruple for matrices with arbitrary nonzero upper-right entries.

Lemma 2. *Let $N_1 = \begin{pmatrix} u & 1 \\ 0 & v \end{pmatrix} \in UT_2(\mathbb{Z})$ and assume that $\{A_1, A_2, A_3, A_4\}$ is an upper-triangular Jordan $D(N_1)$ -quadruple in $UT_2(\mathbb{Z})$, where*

$$A_i = \begin{pmatrix} a_i & b_i \\ 0 & c_i \end{pmatrix}, \quad 1 \leq i \leq 4.$$

Then for every $m \in \mathbb{Z} \setminus \{0\}$, the matrices

$$A_i^{(m)} = \begin{pmatrix} a_i & mb_i \\ 0 & c_i \end{pmatrix}, \quad 1 \leq i \leq 4$$

form an upper-triangular Jordan $D(N_m)$ -quadruple for $N_m = \begin{pmatrix} u & m \\ 0 & v \end{pmatrix}$.

Proof. Assume that

$$A_i \circ A_j + N_1 = R_{ij}^2, \quad R_{ij} = \begin{pmatrix} r_{ij} & z_{ij} \\ 0 & t_{ij} \end{pmatrix},$$

for each pair $1 \leq i < j \leq 4$. Since,

$$[A_i^{(m)} \circ A_j^{(m)} + N_m]_{11} = r_{ij}^2, \quad [A_i^{(m)} \circ A_j^{(m)} + N_m]_{22} = t_{ij}^2$$

and

$$[A_i^{(m)} \circ A_j^{(m)} + N_m]_{12} = m[A_i \circ A_j + N_1]_{12} = mz_{ij}(r_{ij} + t_{ij}),$$

i.e. since the upper-right entry in the Jordan product and in the square of an upper-triangular matrix depends linearly on the upper-right entries when the diagonals are fixed, one has

$$A_i^{(m)} \circ A_j^{(m)} + N_m = \left(R_{ij}^{(m)}\right)^2, \quad R_{ij}^{(m)} = \begin{pmatrix} r_{ij} & mz_{ij} \\ 0 & t_{ij} \end{pmatrix}.$$

Hence $\{A_1^{(m)}, A_2^{(m)}, A_3^{(m)}, A_4^{(m)}\}$ is a Jordan $D(N_m)$ -quadruple. $\square$

Example 4. Let $N = \begin{pmatrix} 4 & 2 \\ 0 & 4 \end{pmatrix}$. The set

$$\{A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 2 \\ 0 & 12 \end{pmatrix}, C = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}, D = \begin{pmatrix} 7 & 1 \\ 0 & 5 \end{pmatrix}\}$$

is a $D(N)$ -quadruple under the Jordan product. Indeed,

$$A \circ B + N = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}^2, A \circ C + N = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}^2, A \circ D + N = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}^2,$$

$$B \circ C + N = \begin{pmatrix} 2 & 3 \\ 0 & 4 \end{pmatrix}^2, B \circ D + N = \begin{pmatrix} 2 & 2 \\ 0 & 8 \end{pmatrix}^2, C \circ D + N = \begin{pmatrix} 5 & 2 \\ 0 & 3 \end{pmatrix}^2.$$

Although N is not a difference of two squares in $UT_2(\mathbb{Z})$ (according to [11]), it has several such representations in $M_2(\mathbb{Z})$:

$$\begin{aligned} N &= \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix}^2 - \begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix}^2 - \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}^2 - \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}^2 = \\ &= \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}^2 - \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}^2 = \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}^2 - \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^2. \end{aligned}$$

This example shows that the existence of Jordan $D(N)$ -quadruples does not imply representability of N as a difference of two squares in $UT_2(\mathbb{Z})$.

3.2. Matrices representable as differences of two squares. In many commutative rings, it has been shown that a $D(n)$ -quadruple exists if and only if n can be represented as a difference of two squares, up to a finite number of exceptions. Although we do not expect such a thing in matrix structures, we believe that it makes sense to explore the connection between these two concepts. In [11] we gave a complete characterization of upper-triangular matrices that can be represented as a difference of two squares in terms of congruences modulo 16.

We first show that every difference-of-squares representation naturally yields a Jordan $D(N)$ -quadruple. Let us start with a *degenerate construction* of Jordan $D(N)$ -quadruple.

Lemma 3 (Degenerate construction). *Let $N = K^2 - A_0^2 \in M_n(\mathbb{Z})$. The family*

$$(A_0, A_0, 2A_0 + 2K, 5A_0 + 4K)$$

has the Jordan $D(N)$ -property.

Proof. $\{A_0, A_0, 2A_0 + 2K\}$ and $\{A_0, 2A_0 + 2K, 5A_0 + 4K\}$ are regular $D(N)$ -triples. Indeed,

$$A_0 \circ A_0 + N = K^2 \quad \text{and} \quad A_0 + A_0 + K = 2A_0 + 2K,$$

$$A_0 \circ (2A_0 + 2K) + N = 2A_0^2 + 2A_0 \circ K + K^2 - A_0^2 = (A_0 + K)^2$$

and

$$A_0 + (2A_0 + 2K) + 2(A_0 + K) = 5A_0 + 4K.$$

□

We will show the result on converting a degenerate quadruple arising from a difference of squares representation into infinitely many genuine Jordan $D(N)$ -quadruples. This result requires the following technical lemma.

Lemma 4. *If $N \in UT_2(\mathbb{Z})$ can be represented as a difference of two squares in $UT_2(\mathbb{Z})$, then there exist $A_0, K \in UT_2(\mathbb{Z})$ such that*

- (i) $N = K^2 - A_0^2$,
- (ii) $3A_0 + 2K \neq 0$, $A_0 + K \neq 0$ and $5A_0 + 4K \neq 0$,
- (iii) either $\text{tr}(2A_0 + K) \neq 0$, or $\text{tr}(A_0) = \text{tr}(K) = 0$.

Proof. Assume first that $N = 0$. Then we may take $A_0 = K = I$, so all three conditions are clearly satisfied.

Now let $N \neq 0$, and suppose that $N = K^2 - A_0^2$ for some $A_0, K \in UT_2(\mathbb{Z})$. Note that $A_0 + K \neq 0$ and $-A_0 + K \neq 0$, since either equality would imply $K^2 = A_0^2$ and hence $N = 0$. We distinguish two cases: $A_0 = 0$ and $A_0 \neq 0$.

Case $A_0 = 0$: Note that $K \neq 0$, since $N \neq 0$. Hence, $3A_0 + 2K \neq 0$ and $5A_0 + 4K \neq 0$. Moreover, either $\text{tr} K \neq 0$, in which case $\text{tr}(2A_0 + K) \neq 0$, or $\text{tr} K = 0$, in which case $\text{tr}(A_0) = \text{tr}(K) = 0$.

Case $A_0 \neq 0$: We analyze the possible obstructions to conditions (ii) and (iii) and, if necessary, fix them by replacing A_0 by $-A_0$.

- ▷ If $3A_0 + 2K = 0$, then $-3A_0 + 2K \neq 0$, since if both equalities hold, then $A_0 = K = 0$. In addition, $-5A_0 + 4K \neq 0$, because $3A_0 + 2K = 0$ and $-5A_0 + 4K = 0$ yield $A_0 = K = 0$. Hence after replacing A_0 by $-A_0$, conditions (i) and (ii) are satisfied.

If $\text{tr}(-2A_0 + K) = 0$, then together with $3A_0 + 2K = 0$ it follows that $\text{tr}(-A_0) = \text{tr}(K) = 0$. Hence, by replacing A_0 by $-A_0$, we meet all three conditions.

- ▷ If $5A_0 + 4K = 0$, by an analogous argument as in the previous case, all three conditions are satisfied by replacing A_0 by $-A_0$.
- ▷ If $\text{tr}(2A_0 + K) = 0$ and $\text{tr} A_0 \neq 0$, $\text{tr} K \neq 0$. Note that $\text{tr}(-2A_0 + K) \neq 0$, since otherwise both $\text{tr}(2A_0 + K) = 0$ and $\text{tr}(-2A_0 + K) = 0$ would hold, implying $\text{tr} A_0 = \text{tr} K = 0$. Also, if $-3A_0 + 2K = 0$, then $\text{tr}(2A_0 + K) = 0$ implies $\text{tr} A_0 = \text{tr} K = 0$. Hence, $-3A_0 + 2K \neq 0$. By the same argument we get $-5A_0 + 4K \neq 0$. Therefore, replacing A_0 by $-A_0$ satisfies both (ii) and (iii).

□

Theorem 5 (Prototype quadruple). *Let $A_0, K \in UT_2(\mathbb{Z})$ be chosen as in Lemma 4, and let*

$$N = K^2 - A_0^2, \quad U_t = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}, \quad t \in \mathbb{Z}.$$

Then there exist infinitely many $t \in \mathbb{Z} \setminus \{0\}$ such that

$$(4) \quad \{A_0 + U_t, A_0 - U_t, 2A_0 + 2K, 5A_0 + 4K\}$$

is a Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z}) \setminus \{0\}$.

Proof. First note that $(2A_0 + 2K) \circ (5A_0 + 4K) + N = \square$, since $\{A_0, 2A_0 + 2K, 5A_0 + 4K\}$ is a Jordan $D(N)$ -triple. Moreover, U_t is a nilpotent matrix, i.e. $U_t^2 = 0$. Thus, it remains to verify the five remaining pairwise Jordan products.

$$\begin{aligned} (1) \quad & (A_0 + U_t) \circ (A_0 - U_t) + N = A_0^2 + N = K^2 \\ (2) \quad & (A_0 + U_t) \circ (2A_0 + 2K) + N = (A_0 + K + U_t)^2 \\ (3) \quad & (A_0 + U_t) \circ (5A_0 + 4K) + N = (2A_0 + K + U_t)^2 + \frac{1}{2} \begin{pmatrix} 0 & t \cdot \text{tr}(A_0 + 2K) \\ 0 & 0 \end{pmatrix} \\ (4) \quad & (A_0 - U_t) \circ (2A_0 + 2K) + N = (A_0 + K - U_t)^2 \\ (5) \quad & (A_0 - U_t) \circ (5A_0 + 4K) + N = (2A_0 + K - U_t)^2 - \frac{1}{2} \begin{pmatrix} 0 & t \cdot \text{tr}(A_0 + 2K) \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Conditions (1), (2), and (4) are immediate. It remains to consider (3) and (5), i.e. to find t such that the matrices on the RHS are perfect squares. By formula (3), if $x + z \neq 0$, a matrix $\begin{pmatrix} x^2 & y \\ 0 & z^2 \end{pmatrix}$ is a square if and only if its upper-right entry is divisible by $x + z$, i.e. by the trace of the square root. If $x + z = 0$, such a square exists if and only if $y = 0$. For the matrices appearing in (3) and (5), the candidate square roots are $2A_0 + K \pm U_t$, whose traces equal

$$\text{tr}(2A_0 + K \pm U_t) = \text{tr}(2A_0 + K),$$

since U_t has trace zero. The correction terms in (3) and (5) contribute upper-right entries equal to

$$\pm \frac{t}{2} \text{tr}(A_0 + 2K).$$

Therefore it suffices to choose t so that these entries are divisible by $\text{tr}(2A_0 + K)$. By Lemma 4 there are two possibilities: either $\text{tr}(2A_0 + K) \neq 0$, or $\text{tr} A_0 = \text{tr} K = 0$.

If $\text{tr}(2A_0 + K) \neq 0$, then matrices in (3) and (5) are perfect squares for

$$t = 2\alpha \text{tr}(2A_0 + K), \quad \alpha \in \mathbb{Z} \setminus \{0\}.$$

If $\text{tr} A_0 = \text{tr} K = 0$, then also $\text{tr}(2A_0 + K) = 0$. Hence the correction terms in (3) and (5) vanish, and both identities reduce to perfect squares for every $t \neq 0$.

It remains to verify that the four matrices in (4) are distinct. Denote the four matrices in (4) with B_1 to B_4 .

- $B_1 = B_2 \iff U_t = 0$
But $U_t \neq 0$ since $t \neq 0$.
- $B_1 = B_3 \iff U_t = A_0 + 2K$
The identity $U_t = A_0 + 2K$ is possible for at most one value of t .
The same argument applies to the following three cases:
- $B_1 = B_4 \iff U_t = 4A_0 + 4K$
- $B_2 = B_3 \iff U_t = -A_0 - 2K$

- $B_2 = B_4 \iff U_t = -4A_0 - 4K$
- $B_3 = B_4 \iff 3A_0 + 2K = 0$.

But $3A_0 + 2K \neq 0$ by Lemma 4.

Hence only finitely many values of t must be excluded. Since the construction provides infinitely many admissible values of t , infinitely many remain for which the four matrices are pairwise distinct.

Moreover, after excluding finitely many values of t , we also have $B_i \neq 0$ for all $i = 1, 2, 3, 4$. Indeed, Lemma 4 gives $A_0 + K \neq 0$ and $5A_0 + 4K \neq 0$, hence $2A_0 + 2K \neq 0$ and $5A_0 + 4K \neq 0$. The equalities $A_0 + U_t = 0$ and $A_0 - U_t = 0$ can occur for at most one value of t each, so these values can be excluded together with the finitely many values producing coincidences. $\square$

Remark 1. *In the construction of Theorem 5, suppose in addition that*

$$A_0^2 \neq 0, \quad \text{tr}(A_0 + K) \neq 0, \quad \text{tr}(5A_0 + 4K) \neq 0, \quad (A_0 + K) \circ (5A_0 + 4K) \neq 0.$$

*Then infinitely many of the constructed Jordan $D(N)$ -quadruples are **non-degenerate**, i.e. all their pairwise Jordan products are nonzero. Indeed, the two fixed products are*

$$(A_0 + U_t) \circ (A_0 - U_t) = A_0^2$$

and

$$(2A_0 + 2K) \circ (5A_0 + 4K) = 2(A_0 + K) \circ (5A_0 + 4K)$$

which are nonzero by assumption. The remaining four pairwise Jordan products $B_1 \circ B_3$, $B_2 \circ B_3$, $B_1 \circ B_4$ and $B_2 \circ B_4$ depend linearly on t . Their upper-right entries are affine linear functions of t , with coefficients $\text{tr}(A_0 + K)$ and $\frac{1}{2} \text{tr}(5A_0 + 4K)$, respectively. Since these coefficients are nonzero by assumption, each square condition excludes at most one value of t . Excluding finitely many such values, infinitely many admissible values of t remain.

Although not every representation $N = K^2 - A_0^2$ is suitable for the construction of Theorem 5, Lemma 4 guarantees the existence of a favourable representation. Hence every matrix $N \in UT_2(\mathbb{Z})$ representable as a difference of two squares gives rise to infinitely many Jordan $D(N)$ -quadruples. Combining this with the characterization of upper-triangular 2×2 integer matrices representable as differences of two squares, given in [11], we obtain the following corollary.

Corollary 6. *Let*

$$N = \begin{pmatrix} p & r \\ 0 & q \end{pmatrix} \in UT_2(\mathbb{Z}).$$

If one of the following holds:

- (1) *p and q are odd, and*
 - $p \equiv q \pmod{4}$ and $r \equiv 0 \pmod{2}$;
 - $p \not\equiv q \pmod{4}$ and r is arbitrary;
- (2) *one of p, q is odd and the other is divisible by 4, and r is arbitrary;*
- (3) *$p \equiv q \equiv 0 \pmod{4}$, and*

- $(p, q) \bmod 16 \in \{(4, 4), (12, 12)\}$ and $r \equiv 0 \pmod{4}$;
- $(p, q) \bmod 16 \in \{(4, 12), (8, 8), (12, 4)\}$ and $r \equiv 0 \pmod{2}$;
- otherwise, r is arbitrary,

then there exist infinitely many Jordan $D(N)$ -quadruples in $UT_2(\mathbb{Z})$.

The remaining congruence classes therefore correspond precisely to matrices that are not representable as differences of two squares in $UT_2(\mathbb{Z})$.

4. MATRICES NOT REPRESENTABLE AS DIFFERENCES OF TWO SQUARES

According to Corollary 6, it remains to investigate matrices that are not representable as differences of two squares in $UT_2(\mathbb{Z})$, i.e. for the following matrix forms:

- (I) $p \equiv 2 \pmod{4}$ or $q \equiv 2 \pmod{4}$,
- (II) $p \equiv q \pmod{4}$ and $r \equiv 1 \pmod{2}$,
- (III) $(p, q) \equiv (4, 4), (12, 12) \pmod{16}$, $r \equiv 1, 2, 3 \pmod{4}$,
- (IV) $(p, q) \equiv (4, 12), (8, 8), (12, 4) \pmod{16}$, $r \equiv 1 \pmod{2}$.

Throughout this section, let

$$(5) \quad A_i = \begin{pmatrix} a_i & b_i \\ 0 & d_i \end{pmatrix}, \quad i = 1, \dots, 4,$$

be a Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$, where $N = \begin{pmatrix} p & r \\ 0 & q \end{pmatrix}$. Then, for every $1 \leq i < j \leq 4$,

$$(6) \quad A_i \circ A_j + N = \begin{pmatrix} a_i a_j + p & b_i c_j + b_j c_i + r \\ 0 & d_i d_j + q \end{pmatrix}$$

is a square in $UT_2(\mathbb{Z})$, where we denoted

$$(7) \quad c_k = \frac{a_k + d_k}{2} = \frac{1}{2} \operatorname{tr} A_k, \quad k = 1, 2, 3, 4.$$

The proofs of the non-existence results rely on the following three ingredients:

(i) **Scalar $D(n)$ -conditions**

The diagonal entries of a Jordan $D(N)$ -quadruple form classical $D(p)$ and $D(q)$ -quadruples. Consequently, known congruence restrictions for integer $D(n)$ -quadruples impose parity conditions on the diagonal entries of matrices A_i 's.

(ii) **Integrality condition**

Since the Jordan product of two upper-triangular integer matrices must again have integer entries, its upper-right entry satisfies additional integrality (and consequently parity) constraints.

(iii) **Square condition**

Finally, if $A_i \circ A_j + N$ is a square in $UT_2(\mathbb{Z})$, formula (6) imposes divisibility conditions on the upper-right entry. Precisely, it implies that its upper-right entry is divisible by the trace of its square root.

The interplay between these three ingredients yields the desired contradictions.

4.1. **Case I.** $p \equiv 2 \pmod{4}$ or $q \equiv 2 \pmod{4}$.

Proposition 7. *Let*

$$N = \begin{pmatrix} p & m \\ 0 & q \end{pmatrix} \in UT_2(\mathbb{Z}).$$

If $p \equiv 2 \pmod{4}$ or $q \equiv 2 \pmod{4}$, then there is no upper-triangular Jordan $D(N)$ -quadruple.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions, the diagonal entries a_1, a_2, a_3, a_4 form a classical $D(p)$ -quadruple, whereas d_1, d_2, d_3, d_4 form a classical $D(q)$ -quadruple. Since no four integers have the $D(n)$ -property when $n \equiv 2 \pmod{4}$ (according to Thm. 1 in [1]), the conclusion follows immediately. $\square$

4.2. **Case II.** $p \equiv q \pmod{4}$ and $r \equiv 1 \pmod{2}$.

Proposition 8. *Let*

$$N = \begin{pmatrix} 4k+3 & m \\ 0 & 4n+3 \end{pmatrix}, \quad m \equiv 1 \pmod{2}.$$

Then there is no upper-triangular Jordan $D(N)$ -quadruple.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions, for every $i \neq j$, the diagonal entries of $A_i \circ A_j + N$ satisfy

$$a_i a_j + (4k+3) = \square, \quad d_i d_j + (4n+3) = \square.$$

Hence, among a_1, a_2, a_3, a_4 there is at most one even element, and the same holds for d_1, d_2, d_3, d_4 . Therefore the following parity patterns are possible:

- (a) all a_i and d_i are odd,
- (b) exactly one a_i is even, all d_i are odd,
- (c) exactly one d_i is even, all a_i are odd,
- (d) exactly one a_i is even and exactly one d_i is even.

Case (a): Since $a_i + d_i$ is even, so $c_i \in \mathbb{Z}$ for $i = 1, 2, 3, 4$. Therefore, the upper-right element of $A_i \circ A_j + N$ is an integer:

$$[A_i \circ A_j + N]_{12} = b_i c_j + b_j c_i + m,$$

for all $1 \leq i < j \leq 4$. Since the diagonal entries of $A_i \circ A_j + N$ are divisible by 4, the square condition implies that its upper-right entry is even. As m is odd, (6) gives

$$b_i c_j + b_j c_i \equiv 1 \pmod{2}, \quad 1 \leq i < j \leq 4,$$

which is not possible. Indeed, let $v_i = (b_i, c_i) \pmod{2} \in \mathbb{Z}_2^2$. Note that $\det \begin{pmatrix} v_i \\ v_j \end{pmatrix} = 1$ if and only if $v_i \neq v_j$ and both v_i and v_j are not $(0, 0)$, but there are only three nonzero vectors in $\mathbb{Z}_2^2$.

Case (b): Assume without loss of generality that a_1 is even. Then

$$a_1 + d_1 \equiv 1 \pmod{2}, \quad a_i + d_i \equiv 0 \pmod{2}, \quad c_i \in \mathbb{Z}, \quad i = 2, 3, 4.$$

By the integrality condition applied to $A_1 \circ A_j$, $j = 2, 3, 4$, we get

$$b_j(a_1 + d_1) + b_1(a_j + d_j) \equiv 0 \pmod{2}.$$

Therefore b_j is even for $j = 2, 3, 4$. Hence

$$[A_i \circ A_j]_{12} = b_i c_j + b_j c_i \equiv 0 \pmod{2}, \quad 2 \leq i < j \leq 4.$$

On the other hand, for $i, j \in \{2, 3, 4\}$, all a_i, d_i are odd. Thus, by the same argument as in Case (a), the square condition gives

$$[A_i \circ A_j]_{12} \equiv 1 \pmod{2}, \quad 2 \leq i < j \leq 4,$$

a contradiction.

Case (c): Analogous to Case (b).

Case (d): There are two subcases: if even diagonal elements occur in the same matrix or if they occur in different matrices.

If the even entries occur in the same matrix, say a_1 and d_1 , then $a_i + d_i$ is even for all i , so the argument of Case (a) applies and gives a contradiction.

If the even entries occur in different matrices, say a_1 and d_2 are even, then for $j = 3, 4$, the integrality condition applied to $A_1 \circ A_j$ and $A_2 \circ A_j$ forces b_3 and b_4 to be even. Hence $[A_3 \circ A_4]_{12}$ is even. But a_3, a_4, d_3, d_4 are all odd, so the square condition, as in case (a), forces $[A_3 \circ A_4]_{12}$ to be odd, a contradiction. $\square$

The previous proposition shows that Jordan $D(N)$ -quadruples do not exist for one of the congruence classes arising in Case II (" $4k + 3 \& 4n + 3$ "). We now show that the complementary congruence (" $4k + 1 \& 4n + 1$ ") class behaves differently by constructing explicit quadruples.

Proposition 9. *If*

$$N = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}, \quad m \in \mathbb{Z},$$

then the matrices

$$A_1 = \begin{pmatrix} 1 & -24m \\ 0 & 8 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 3 & 12m \\ 0 & 120 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 8 & -2m \\ 0 & 3 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 120 & 6m \\ 0 & 1 \end{pmatrix}.$$

form an upper-triangular Jordan $D(N)$ -quadruple consisting of invertible matrices.

Remark 2. *Note that the top and bottom diagonal elements are the two permutations $(1, 3, 8, 120)$ and $(8, 120, 3, 1)$ of a scalar $D(1)$ -quadruple, i.e. Fermat's quadruple $\{1, 3, 8, 120\}$.*

Proof. A routine computation yields

$$A_1 \circ A_2 + N = \begin{pmatrix} 4 & -1421m \\ 0 & 961 \end{pmatrix} = \begin{pmatrix} 2 & 49m \\ 0 & -31 \end{pmatrix}^2,$$

$$\begin{aligned}
A_1 \circ A_3 + N &= \begin{pmatrix} 9 & -140m \\ 0 & 25 \end{pmatrix} = \begin{pmatrix} -3 & -70m \\ 0 & 5 \end{pmatrix}^2, \\
A_1 \circ A_4 + N &= \begin{pmatrix} 121 & -1424m \\ 0 & 9 \end{pmatrix} = \begin{pmatrix} 11 & -178m \\ 0 & -3 \end{pmatrix}^2, \\
A_2 \circ A_3 + N &= \begin{pmatrix} 25 & -56m \\ 0 & 361 \end{pmatrix} = \begin{pmatrix} -5 & -4m \\ 0 & 19 \end{pmatrix}^2, \\
A_2 \circ A_4 + N &= \begin{pmatrix} 361 & 1096m \\ 0 & 121 \end{pmatrix} = \begin{pmatrix} 19 & 137m \\ 0 & -11 \end{pmatrix}^2, \\
A_3 \circ A_4 + N &= \begin{pmatrix} 961 & -87m \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -31 & 3m \\ 0 & 2 \end{pmatrix}^2.
\end{aligned}$$

□

Proposition 10. For $N = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, the five matrices

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 3 & 2 \\ 0 & 0 \end{pmatrix}$$

form a Jordan $D(N)$ -quintuple. Moreover, this quintuple cannot be extended to a Jordan $D(N)$ -sixture by any upper-triangular integer matrix.

Proof. Directly,

$$\begin{aligned}
A \circ B + N &= \begin{pmatrix} 1 & 1 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 0 & 2 \end{pmatrix}^2, \\
A \circ C + N &= A \circ D + N = A \circ E + N = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2, \\
B \circ C + N &= B \circ D + N = B \circ E + N = C \circ D + N = C \circ E + N = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}^2, \\
D \circ E + N &= \begin{pmatrix} 4 & 5 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 0 & -1 \end{pmatrix}^2.
\end{aligned}$$

So the five matrices form a Jordan $D(N)$ -quintuple.

Now suppose $F = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ extends it to a Jordan $D(N)$ -sixture. From the pairs (F, D) and (F, E) one gets that $a + 1$ and $3a + 1$ must both be squares, hence a is even. Similarly, from (F, A) and (F, B) one gets that $d + 1$ and $3d + 1$ are squares, hence d is even. But then

$$C \circ F + N = \begin{pmatrix} 1 & a + d + 1 \\ 0 & 1 \end{pmatrix}$$

has odd upper-right entry, while by the form (3), a square with diagonal entries 1, 1 must have even upper-right entry. Contradiction. □

Remark 3. Note that N from Proposition 9 can be represented as a difference of two squares in $UT_2(\mathbb{Z})$ if and only if m is even. Thus, for odd m , Proposition 9 provides Jordan $D(N)$ -quadruples for matrices that are not representable as differences of two squares. This indicates that the " $4k+1$ & $4n+1$ case" will be quite different.

Proposition 11. Let

$$N = \begin{pmatrix} 8k+1 & 2m+1 \\ 0 & 8n+5 \end{pmatrix} \quad \text{or} \quad N = \begin{pmatrix} 8k+5 & 2m+1 \\ 0 & 8n+1 \end{pmatrix} \in UT_2(\mathbb{Z}).$$

Then there is no upper-triangular Jordan $D(N)$ -quadruple.

Proof. It suffices to consider

$$N = \begin{pmatrix} 8k+1 & 2m+1 \\ 0 & 8n+5 \end{pmatrix},$$

since the other case is symmetric. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions, the top diagonal entries satisfy

$$a_i a_j + 1 \equiv \square \pmod{8}, \quad i \neq j.$$

Thus, up to permutation, the possible residue multisets $\{a_1, a_2, a_3, a_4\}$ modulo 8 are:

$$\begin{aligned} & (0, 0, 0, r), r \in \mathbb{Z}_8, \\ & (0, 0, 1, 3), (0, 0, 1, 7), (0, 0, 2, 4), (0, 0, 3, 5), (0, 0, 4, 4), (0, 0, 4, 6), (0, 0, 5, 7), \\ & (0, 2, 4, 4), (0, 4, 4, 4), (0, 4, 4, 6), (2, 4, 4, 4), (4, 4, 4, 4), (4, 4, 4, 6). \end{aligned}$$

Similarly, the bottom diagonal entries satisfy

$$d_i d_j + 5 \equiv \square \pmod{8}, \quad i \neq j,$$

and hence, up to permutation, the possible residue multisets $\{d_1, d_2, d_3, d_4\}$ modulo 8 are more restrictive:

$$(2, 2, 2, 2), (2, 2, 2, 6), (2, 2, 6, 6), (2, 6, 6, 6), (6, 6, 6, 6).$$

In particular, all $d_i \equiv 2 \pmod{4}$.

We consider, up to permutation, the following cases:

- (a) all a_i are even,
- (b) $(a_1, a_2, a_3, a_4) \equiv (0, 0, 0, r) \pmod{8}$, r is odd,
- (c) $(a_1, a_2, a_3, a_4) \equiv (0, 0, 1, 3), (0, 0, 1, 7), (0, 0, 3, 5), (0, 0, 5, 7) \pmod{8}$.

Case (a): Since all d_i are even, all $a_i + d_i$ are even and $c_i \in \mathbb{Z}$, $i = 1, 2, 3, 4$. Moreover, both diagonal entries of $A_i \circ A_j + N$ are odd, so the square condition implies that the upper-right entry of $A_i \circ A_j + N$ is even. Since $2m+1$ is odd, (6) gives

$$b_i c_j + b_j c_i \equiv 1 \pmod{2}, \quad 1 \leq i < j \leq 4.$$

This is impossible by the same argument as in Proposition 8, Case (a).

Case (b): Assume

$$a_1 \equiv a_2 \equiv a_3 \equiv 0 \pmod{8}, \quad a_4 \equiv 1 \pmod{2}.$$

Since $d_i \equiv 2 \pmod{4}$, $i = 1, 2, 3, 4$, we have

$$a_i + d_i \equiv 2 \pmod{4}, \quad i = 1, 2, 3, \quad a_4 + d_4 \equiv 1 \pmod{2}.$$

By the integrality condition applied to $A_i \circ A_4$, $i = 1, 2, 3$, it follows that b_1, b_2, b_3 are even. Now consider $A_1 \circ A_2 + N$. Its diagonal entries are odd, so the square condition forces its upper-right entry to be even. On the other hand, using (6) and the fact that b_1, b_2 are even, we obtain

$$[A_1 \circ A_2 + N]_{12} = b_1 c_2 + b_2 c_1 + 2m + 1 \equiv 1 \pmod{2},$$

a contradiction.

Case (c): We may assume

$$a_1 \equiv a_2 \equiv 0 \pmod{8}, \quad a_3 \equiv a_4 \equiv 1 \pmod{2}.$$

Again,

$$a_1 + d_1 \equiv a_2 + d_2 \equiv 2 \pmod{4},$$

while

$$a_3 + d_3 \equiv a_4 + d_4 \equiv 1 \pmod{2}.$$

By the integrality condition applied to $A_1 \circ A_3$ and $A_1 \circ A_4$, we get $b_1 \equiv 0 \pmod{2}$. Similarly, integrality of $A_2 \circ A_3$ and $A_2 \circ A_4$ forces $b_2 \equiv 0 \pmod{2}$. Hence,

$$[A_1 \circ A_2 + N]_{12} = b_1 c_2 + b_2 c_1 + 2m + 1 \equiv 1 \pmod{2}.$$

But the diagonal entries of $A_1 \circ A_2 + N$ are odd, so the square condition requires its upper-right entry to be even, a contradiction.

Thus all possible residue patterns are impossible, and hence no Jordan $D(N)$ -quadruple exists. $\square$

Proposition 12. Let $N = \begin{pmatrix} 8k+5 & 2m+1 \\ 0 & 8n+5 \end{pmatrix} \in UT_2(\mathbb{Z})$. Then there is no upper-triangular Jordan $D(N)$ -quadruple.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions,

$$a_i a_j + 8k + 5 = \square, \quad d_i d_j + 8n + 5 = \square, \quad 1 \leq i < j \leq 4.$$

Reducing modulo 8, this implies

$$a_i, d_i \equiv 2 \text{ or } 6 \pmod{8}, \quad i = 1, 2, 3, 4.$$

Hence all a_i and d_i are even. Since $[N]_{12} = 2m + 1$ is odd, we are exactly in Case (a) of the proof of Proposition 11, which gives a contradiction. $\square$

Proposition 13. Let $N = \begin{pmatrix} s^2 & m \\ 0 & s^2 \end{pmatrix}$. For every $s, m \in \mathbb{Z}$, there exists a Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$.

Proof. Matrices

$$A_1 = \begin{pmatrix} 0 & -2m \\ 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

$$A_3 = \begin{pmatrix} 2s+1 & 0 \\ 0 & 4s+4 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 4s+4 & -2m \\ 0 & 2s+1 \end{pmatrix}$$

form a Jordan $D(N)$ -quadruple. Indeed, a direct computation gives

$$A_1 \circ A_2 + N = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix}^2, \quad A_1 \circ A_3 + N = \begin{pmatrix} s & m(3s+2) \\ 0 & -(s+2) \end{pmatrix}^2,$$

$$A_1 \circ A_4 + N = \begin{pmatrix} s & m(6s+5) \\ 0 & -(s+1) \end{pmatrix}^2, \quad A_2 \circ A_3 + N = \begin{pmatrix} s+1 & m \\ 0 & -s \end{pmatrix}^2,$$

$$A_2 \circ A_4 + N = \begin{pmatrix} s+2 & 0 \\ 0 & -s \end{pmatrix}^2, \quad A_3 \circ A_4 + N = \begin{pmatrix} 3s+2 & -m \\ 0 & 3s+2 \end{pmatrix}^2.$$

□

Remark 4. If $s^2 \equiv 1 \pmod{8}$ and $m \equiv 1 \pmod{2}$ then the previous proposition shows that there are infinitely many N of the form $\begin{pmatrix} 8k+1 & m \\ 0 & 8k+1 \end{pmatrix}$ such that a Jordan $D(N)$ -quadruple exists, and N cannot be represented as a difference of two squares.

4.3. Case III. $(p, q) \equiv (4, 4), (12, 12) \pmod{16}$, $r \equiv 1, 2, 3 \pmod{4}$.

Proposition 14. Let $m \in \mathbb{Z}$, $4 \nmid m$ and

$$N = \begin{pmatrix} 16k+12 & m \\ 0 & 16n+12 \end{pmatrix} \in UT_2(\mathbb{Z}).$$

Then there is no Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions, for $i \neq j$

$$a_i a_j + 16k + 12 = \square, \quad d_i d_j + 16n + 12 = \square.$$

Since the quadratic residues modulo 16 are 0, 1, 4, 9, a direct check modulo 16 shows that, up to permutation,

$$(a_1, a_2, a_3, a_4) \equiv (2, 2, 2, 2) \quad \text{or} \quad (0, 2, 2, 2) \pmod{4}.$$

The same conclusion holds for (d_1, d_2, d_3, d_4) . In particular, $a_i + d_i$ is even for every i , so $c_i = (a_i + d_i)/2 \in \mathbb{Z}$.

Case 1: m is odd

Note that we are in the same situation as Case (a) of Proposition 8. Indeed, all diagonal entries of $A_i \circ A_j + N$ are even squares, so its upper-right entry is even. Since m is odd, this forces $c_i b_j + c_j b_i$ to be odd for every $i < j$, and the same $\mathbb{Z}_2^2$ -argument gives a contradiction.

Case 2: $m \equiv 2 \pmod{4}$

Up to permutations of the indices and symmetry between the top and bottom diagonal entries, it suffices to consider the following four subcases.

Subcases:

$$\begin{array}{ll}
 \mathbf{2a)} & \begin{array}{l} a_i \pmod 4 : 2 \ 2 \ 2 \ 2 \\ d_i \pmod 4 : 2 \ 2 \ 2 \ 2 \end{array} \\
 \mathbf{2b)} & \begin{array}{l} a_i \pmod 4 : 2 \ 2 \ 2 \ 2 \\ d_i \pmod 4 : 2 \ 2 \ 2 \ 0 \end{array} \\
 \mathbf{2c)} & \begin{array}{l} a_i \pmod 4 : 2 \ 2 \ 2 \ 0 \\ d_i \pmod 4 : 2 \ 2 \ 2 \ 0 \end{array} \\
 \mathbf{2d)} & \begin{array}{l} a_i \pmod 4 : 2 \ 2 \ 2 \ 0 \\ d_i \pmod 4 : 2 \ 2 \ 0 \ 2 \end{array} .
 \end{array}$$

We analyze Subcases 2a) and 2c) simultaneously, and then Subcases 2b) and 2d).

Subcases 2a&2c: In both subcases,

$$a_i + d_i \equiv 0 \pmod{4},$$

and hence

$$c_i \equiv 0 \pmod{2}, \quad i = 1, 2, 3, 4.$$

We claim that, for every pair $i < j$, every upper-triangular square root of $A_i \circ A_j + N$ has trace divisible by 4. In Subcase 2a), for every $i < j$,

$$[A_i \circ A_j + N]_{11} = a_i a_j + 16k + 12 \equiv 4 + 0 + 4 \equiv 0 \pmod{8},$$

and similarly,

$$[A_i \circ A_j + N]_{22} = d_i d_j + 16n + 12 \equiv 0 \pmod{8}.$$

Since these entries are squares, they are in fact congruent to 0 (mod 16). Thus both corresponding diagonal entries of a square root are divisible by 4.

In Subcase 2c), for $1 \leq i < j \leq 3$,

$$[A_i \circ A_j + N]_{11} \equiv [A_i \circ A_j + N]_{22} \equiv 0 \pmod{8},$$

so both diagonal entries are congruent to 0 (mod 16). On the other hand, for $i = 1, 2, 3$,

$$[A_i \circ A_4 + N]_{11} \equiv [A_i \circ A_4 + N]_{22} \equiv 4 \pmod{8}.$$

So, both are congruent to 4 (mod 16), and the corresponding diagonal entries of a square root are both congruent to 2 (mod 4).

Therefore, in every pair, the sum of the diagonal entries of a square root is divisible by 4. Hence the upper-right entry of the square is divisible by 4, and

$$4 \mid [A_i \circ A_j + N]_{12} = c_i b_j + c_j b_i + \underbrace{4\ell + 2}_m.$$

Finally, we obtain

$$c_i b_j + c_j b_i \equiv 2 \pmod{4}, \quad 1 \leq i < j \leq 4.$$

By putting $c_i = 2e_i$, for all i , we get

$$e_i b_j + e_j b_i \equiv 1 \pmod{2}, \quad 1 \leq i < j \leq 4.$$

This is impossible by the same $\mathbb{Z}_2^2$ -argument used in Proposition 8, Case (a).

Subcases 2b&2d: In both subcases, we have

$$c_1 \equiv c_2 \equiv 0 \pmod{2}, \quad c_4 \equiv 1 \pmod{2}.$$

Notice that the parity of c_3 is not needed in the argument. Since diagonal elements of $A_i \circ A_j + N$ are even, the square condition implies that

$$c_i b_j + c_j b_i \equiv 0 \pmod{2}, \quad 1 \leq i < j \leq 4.$$

In particular for $(i, j) = (1, 4)$ and $(2, 4)$, since c_1 and c_2 are even, and c_4 is odd, we get

$$c_1 b_4 + c_4 b_1 \equiv 0 \pmod{2} \implies b_1 \equiv 0 \pmod{2},$$

$$c_2 b_4 + c_4 b_2 \equiv 0 \pmod{2} \implies b_2 \equiv 0 \pmod{2}.$$

Hence,

$$c_1 b_2 + c_2 b_1 \equiv 0 \pmod{4}.$$

For the pair $(1, 2)$, both diagonal entries are congruent to 0 (mod 16), so the diagonal entries of a square root are divisible by 4 and therefore

$$c_1 b_2 + c_2 b_1 \equiv 2 \pmod{4},$$

contradicting the previous congruence. $\square$

Proposition 15. *Let*

$$N = \begin{pmatrix} 16k + 4 & 2m + 1 \\ 0 & 16n + 4 \end{pmatrix} \in UT_2(\mathbb{Z}).$$

Then there is no Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. By the scalar $D(n)$ -conditions, up to permutation, the residues $(a_i \bmod 4)$ and $(d_i \bmod 4)$ are of the forms

$$(0, 0, 0, 0) \text{ or } (0, 0, 0, r) \text{ or } (0, 0, r, r),$$

where $r \in \{1, 2, 3\}$.

We first show that two matrices cannot simultaneously have both diagonal entries divisible by 4. Suppose that there exist two distinct indices $i, j \in \{1, 2, 3, 4\}$ such that

$$a_i \equiv d_i \equiv a_j \equiv d_j \equiv 0 \pmod{4}.$$

Then

$$a_i + d_i \equiv a_j + d_j \equiv 0 \pmod{4} \implies c_i \equiv c_j \equiv 0 \pmod{2}.$$

Therefore,

$$[A_i \circ A_j]_{12} \equiv 0 \pmod{2}.$$

The square condition implies that $[A_i \circ A_j]_{12}$ is odd (since diagonal entries of $A_i \circ A_j + N$ are even and $[N]_{12} = 2m + 1$ is odd). A contradiction.

Hence, only two residue configurations remain:

$$a \equiv (0, 0, r, r) \pmod{4}, \quad d \equiv (0, s, 0, s) \pmod{4},$$

or

$$a \equiv (0, 0, r, r) \pmod{4}, \quad d \equiv (s, s, 0, 0) \pmod{4},$$

with $r, s \in \{1, 2, 3\}$, where $a = (a_1, \dots, a_4)$ and $d = (d_1, \dots, d_4)$.

Case (a): Consider the "crossed type"

$$a \equiv (0, 0, r, r), \quad d \equiv (0, s, 0, s) \pmod{4}.$$

The integrality condition yields

$$sb_1 \equiv rb_1 \equiv (r+s)b_1 \equiv 0 \pmod{2},$$

$$rb_2 + sb_3 \equiv 0 \pmod{2},$$

$$(r+s)b_2 + sb_4 \equiv 0 \pmod{2},$$

$$(r+s)b_3 + rb_4 \equiv 0 \pmod{2}.$$

If r and s have different parity, then $b_1 \equiv b_2 \equiv 0 \pmod{2}$ if s is even and r is odd, or $b_1 \equiv b_3 \equiv 0 \pmod{2}$ if r is even and s is odd. In these two possibilities we obtain a parity obstruction for indices $(1, 2)$ and $(1, 3)$. For instance $[A_i \circ A_j + N]_{12} = \frac{1}{2}b_1s + 2m + 1$ is odd and the square condition forces it to be even because $[A_i \circ A_j + N]_{kk} \equiv 0 \pmod{4}$, $k = 1, 2$.

If $r \equiv s \equiv 2 \pmod{4}$, then $[A_1 \circ A_4]_{12} \equiv \frac{1}{2}b_1(r+s) \pmod{4}$ and so $[A_1 \circ A_4 + N]_{12}$ is odd, while the square condition again forces it to be even, a contradiction.

Finally, suppose that r and s are odd. Then integrality conditions yield

$$b_1 \equiv b_4 \equiv 0 \pmod{2}, \quad b_2 \equiv b_3 \pmod{2}.$$

The square conditions for the pair $(1, 4)$ imply

$$\frac{1}{2}b_1(r+s) + 1 \equiv 0 \pmod{2} \implies b_1(r+s) \equiv 2 \pmod{4}.$$

Since b_1 and $r+s$ are even, the last congruence is impossible. This contradiction eliminates the crossed type.

Case (b): The "disjoint type"

$$a \equiv (0, 0, r, r), \quad d \equiv (s, s, 0, 0) \pmod{4}$$

is treated similarly and again leads to a contradiction.

If r and s are both even, then we have an identical situation as in Case (a) of Proposition 8.

If s is odd then

$$[A_1 \circ A_2 + N]_{22} \equiv s^2 + 4 \equiv 5 \text{ or } 13 \pmod{16}.$$

On the other hand, the square condition gives

$$[A_1 \circ A_2 + N]_{22} \pmod{16} \in \{0, 1, 4, 9\}.$$

A contradiction.

Similarly, if r is odd, then

$$[A_3 \circ A_4 + N]_{11} \equiv r^2 + 4 \equiv 5 \text{ or } 13 \pmod{16}.$$

□

Remark 5. Proposition 13 for $s^2 \equiv 4 \pmod{16}$ and $m \equiv 2 \pmod{4}$ implies the existence of a Jordan $D(N)$ -quadruple for some N of the form $\begin{pmatrix} 16k+4 & 4m+2 \\ 0 & 16n+4 \end{pmatrix}$ which cannot be represented as a difference of two squares in $UT_2(\mathbb{Z})$.

In the next proposition we give a family of Jordan $D(N_t)$ -quadruples for which N_t is representable as a difference of two squares when t is even, but is not representable when t is odd. It follows immediately from Example 4 and Lemma 2 by scaling all upper-right entries by t .

Proposition 16. For every integer $t \neq 0$, the matrices

$$N_t = \begin{pmatrix} 4 & 2t \\ 0 & 4 \end{pmatrix},$$

$$A_t = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}, \quad B_t = \begin{pmatrix} 0 & 2t \\ 0 & 12 \end{pmatrix}, \quad C_t = \begin{pmatrix} 3 & 2t \\ 0 & 1 \end{pmatrix}, \quad D_t = \begin{pmatrix} 7 & t \\ 0 & 5 \end{pmatrix}$$

form an upper-triangular Jordan $D(N_t)$ -quadruple.

4.4. **Case IV.** $(p, q) \equiv (4, 12), (8, 8), (12, 4) \pmod{16}$, $r \equiv 1 \pmod{2}$.

Proposition 17. Let

$$N = \begin{pmatrix} 16k+4 & 2m+1 \\ 0 & 16n+12 \end{pmatrix} \text{ or } N = \begin{pmatrix} 16k+12 & 2m+1 \\ 0 & 16n+4 \end{pmatrix} \in UT_2(\mathbb{Z}).$$

Then there is no Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$.

Proof. Assume that (5) is a Jordan $D(N)$ -quadruple. The scalar $D(n)$ -condition gives

$$(a_1, a_2, a_3, a_4) \equiv (0, 0, 0, 0) \text{ or } (0, 0, 0, r) \text{ or } (0, 0, r, r) \pmod{4}, r \in \{1, 2, 3\},$$

$$(d_1, d_2, d_3, d_4) \equiv (2, 2, 2, 2) \text{ or } (0, 2, 2, 2) \pmod{4}.$$

We have several cases to discuss.

- If all a_i 's are even then we obtain a contradiction in the same manner as in Case (a) of Proposition 8.
- Assume that $a_i \equiv d_i + 2 \equiv 0 \pmod{4}$ for $i = 1, 2, 3$, i.e.

$$\begin{array}{lcl} a_i \pmod{4} & : & 0 \quad 0 \quad 0 \quad * \\ d_i \pmod{4} & : & 2 \quad 2 \quad 2 \quad * \end{array}$$

Define

$$c_i = \frac{1}{2}(a_i + d_i) \equiv 1 \pmod{2}, \quad i = 1, 2, 3.$$

Then for $1 \leq i < j \leq 3$ we have

$$[A_i \circ A_j + N]_{12} = b_i c_j + b_j c_i + 2m + 1 \equiv b_i + b_j + 1.$$

By the square condition (since diagonal elements are even), we get

$$b_1 + b_2 \equiv 1 \pmod{2}, \quad b_1 + b_3 \equiv 1 \pmod{2}, \quad b_2 + b_3 \equiv 1 \pmod{2},$$

which is not possible.

- Assume r is odd and

$$\begin{array}{rcl} a_i \pmod 4 & : & 0 \ 0 \ 0 \ r \\ d_i \pmod 4 & : & 0 \ 2 \ 2 \ 2 \end{array}$$

Hence, with c_i defined as above:

$$c_1 \equiv 0 \pmod 2, \quad c_2 \equiv c_3 \equiv 1 \pmod 2.$$

The square condition (since diagonal elements are even) for the pair of indices (1,2) implies:

$$b_1 \equiv 1 \pmod 2.$$

On the other hand the integrality condition for indices (1,4) forces

$$b_1(a_4 + d_4) + b_4(a_1 + d_1) \equiv b_1 \equiv 0 \pmod 2,$$

a contradiction.

- Assume r is odd and

$$\begin{array}{rcl} a_i \pmod 4 & : & 0 \ 0 \ r \ r \\ d_i \pmod 4 & : & 2 \ 2 \ 2 \ * \end{array}$$

Hence, c_i defined as in the previous case satisfy

$$c_1 \equiv c_2 \equiv 1 \pmod 2.$$

The square condition for the pair of indices (1,2) (since diagonal elements are even) implies:

$$b_1 + b_2 \equiv 1 \pmod 2.$$

The integrality condition for indices (1,3) and (2,3) give:

$$b_1(a_3 + d_3) + b_3(2c_1) \equiv b_1 \equiv 0 \pmod 2$$

$$b_2(a_3 + d_3) + b_3(2c_2) \equiv b_2 \equiv 0 \pmod 2.$$

Therefore, $b_1 + b_2 \equiv 0 \pmod 2$, a contradiction.

- Assume r is odd and

$$\begin{array}{rcl} a_i \pmod 4 & : & 0 \ 0 \ r \ r \\ d_i \pmod 4 & : & 0 \ 2 \ 2 \ 2 \end{array}$$

We have

$$c_1 \equiv 0 \pmod 2, \quad c_2 \equiv 1 \pmod 2.$$

The integrality condition for indices (1,3) gives

$$b_1(a_3 + d_3) + b_3(a_1 + d_1) \equiv b_1 \equiv 0 \pmod 2.$$

The square condition for indices (1,2) implies:

$$b_1c_2 + b_2c_1 + 2m + 1 \equiv 0 \pmod 2,$$

but the LHS is obviously odd, a contradiction.

□

Proposition 18. *Let*

$$N = \begin{pmatrix} 16k + 8 & 2m + 1 \\ 0 & 16n + 8 \end{pmatrix}.$$

Then there is no Jordan $D(N)$ -quadruple in $UT_2(\mathbb{Z})$.

Proof. Since

$$a_i a_j + 8 \equiv \square, \quad d_i d_j + 8 \equiv \square \pmod{16},$$

the only possible residue patterns modulo 4 for a_i 's and d_i 's, up to permutation, are

$$(1, 1, 1, 1), (3, 3, 3, 3), (0, 1, 1, 1), (0, 3, 3, 3).$$

Thus, among the numbers $a_1, \dots, a_4$, at most one is even, and the same holds for $d_1, \dots, d_4$. Consequently, exactly the same four parity cases arise as in Proposition 8. From this point on, the argument depends only on the integrality condition, the square condition for the (1,2)-entries of the Jordan products, and the fact that the (1,2)-entry of N is odd. Therefore, the same parity contradictions as in the proof of Proposition 8 apply without further modification. $\square$

4.5. Summary and remaining cases.

Case	Status	Remaining open cases with
I	completely solved	–
II	partially solved	$(p, q) \equiv (1, 1) \pmod{8}, r \equiv 1 \pmod{2}$
III	partially solved	$(p, q) \equiv (4, 4) \pmod{16}, r \equiv 2 \pmod{4}$
IV	completely solved	–

We emphasize that several infinite families of Jordan $D(N)$ -quadruples have been constructed within the open Cases II and III; see Propositions 9, 13, and 16. Thus, although these cases are not yet completely classified, they already exhibit both existence and non-existence phenomena.

5. POLYNOMIAL CONSTRUCTIONS OF JORDAN $D(N)$ -QUADRUPLES

Although the existence of such quadruples already follows from Corollary 6, we include the following direct construction to illustrate how classical polynomial formulas from [7] for Diophantine quadruples can be adapted to the Jordan setting. The main difference is that, besides matching the diagonal entries, one has to choose the upper-right entries so that the corresponding square conditions are satisfied. In the example below, this requirement reduces to the solvability of a linear Diophantine equation.

We give an explicit construction for

$$N = \begin{pmatrix} 4k + 3 & 2m \\ 0 & 4n + 3 \end{pmatrix}.$$

We begin with the Jordan $D(N)$ -pair

$$\left\{ A = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 9k^2 + 8k + 1 & y \\ 0 & 9n^2 + 8n + 1 \end{pmatrix} \right\}.$$

Note that the diagonal elements are elements of the polynomial quadruple (1) (for $m = 1$). Following the construction of Dujella [7], we extend this pair to a Jordan triple by setting

$$C = A + B + 2R,$$

where

$$A \circ B + N = \begin{pmatrix} (3k+2)^2 & \frac{b}{2}(2+8k+9k^2+8n+9n^2)+y+2m \\ 0 & (3n+2)^2 \end{pmatrix}.$$

In order that $R = \begin{pmatrix} 3k+2 & w \\ 0 & 3n+2 \end{pmatrix}$ satisfy $R^2 = A \circ B + N$, its upper-right entry w must satisfy

$$-\frac{b}{2}(2+8k+9k^2+8n+9n^2) - y + w(4+3k+3n) = 2m.$$

So,

$$y = w(4+3k+3n) - \frac{b}{2}(2+8k+9k^2+8n+9n^2) - 2m,$$

$w, b \in \mathbb{Z}$. If $k+n \equiv 1 \pmod{2}$, then b has to be even. Now, we extend the pair $\{B, C\}$ with

$$D = A + 4B + 4R.$$

Hence, $\{A, B, C, D\}$ is a Jordan $D(N)$ -quadruple if and only if

$$A \circ D + N = \square.$$

We have

$$\frac{1}{2}(AD+DA)+N = \begin{pmatrix} 4(2+3k)^2 & b(14+22k+18k^2+22n+18n^2)+4w+4y+2m \\ 0 & 4(2+3n)^2 \end{pmatrix}.$$

The previous expression equals

$$\begin{pmatrix} 2(2+3k) & z \\ 0 & 2(2+3n) \end{pmatrix}^2$$

if and only if the Diophantine equation

$$b(5+3k+3n) + 2w(5+3k+3n) - z(4+3k+3n) = 3m$$

is solvable in b, w, z . Since

$$\gcd(5+3k+3n, 4+3k+3n) = 1,$$

the equation is solvable. The solution of the previous Diophantine equation is

$$\begin{aligned} z &= (5+3k+3n)p + 3m \\ b &= (4+3k+3n)p + 3m - 2w, \end{aligned}$$

for $p, w \in \mathbb{Z}$. Note that b has to be even if $k+n \equiv 1 \pmod{2}$, but this can be achieved for $p \equiv m \pmod{2}$.

The parameters p and w may be chosen from an infinite arithmetic progression satisfying the required parity condition. Each equality between two of the matrices A, B, C, D , and each condition that one of them is zero, imposes at most one additional linear condition on the free parameters. Excluding these finitely many exceptional choices, we obtain a Jordan $D(N)$ -quadruple of four distinct nonzero matrices.

Together with the prototype construction from Theorem 5, this demonstrates two complementary approaches to constructing Jordan $D(N)$ -quadruples in $UT_2(\mathbb{Z})$: one based on representations as differences of two squares, and one based on adaptations of classical polynomial formulas. The same method applies to other polynomial families of classical $D(n)$ -quadruples as well.

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